Batool hataf jader

مصفوفة ثنائية البعد
2D array
من الصفر حتى الأحترافية

JavaApplication19 - NetBeans IDE 8.2

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help

JavaApplication17.java | JavaApplication18.java | JavaApplication19.java

Source History

```java
package javaapplication19;
public class JavaApplication19 {
public static void main(String[] args) {
    int x[][]={
        {1,2,3},
        {5,6,7},
        {9,8,4},
    };
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++){
    System.out.print(x[i][j]+" ");
    }
    System.out.println();
    }
```

Output - JavaApplication19 (run)

```
run:
1 2 3
5 6 7
9 8 4
BUILD SUCCESSFUL (total time: 0 seconds)
```

طباعة عناصر مصفوفة ثنائية البعد

JavaApplication19 - NetBeans IDE 8.2

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help

JavaApplication17.java | JavaApplication18.java | JavaApplication19.java

```java
public static void main(String[] args) {
    int x[][]={
        {1,2,3},
        {5,6,7},
        {9,8,4},
    };
    int sum=0;
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++){
        sum=sum+x[i][j];
    System.out.print(x[i][j]+" ");
    }
    System.out.println();
    }
    System.out.println("sum:"+sum);
```

Output - JavaApplication19 (run)

```
run:
1 2 3
5 6 7
9 8 4
sum:45
BUILD SUCCESSFUL (total time: 0 seconds)
```

طباعة عناصر مصفوفة ثنائية البعد مع الجمع

JavaApplication19 - NetBeans IDE 8.2

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help

JavaApplication17.java | JavaApplication18.java | JavaApplication19.java

```java
public static void main(String[] args) {
    int x[][]={
        {1,2,3},
        {5,6,7},
        {9,8,4},
    };
    int sum=0;
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++){
        sum=sum+x[i][j];
    System.out.print(x[i][j]+" ");
    }
    System.out.println();
    }
    System.out.println("avarage:"+sum/9);
```

Output - JavaApplication19 (run)

```
run:
1 2 3
5 6 7
9 8 4
avarage:5
BUILD SUCCESSFUL (total time: 0 seconds)
```

طباعة عناصر مصفوفة ثنائية البعد مع المعدل

JavaApplication19 - NetBeans IDE 8.2

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help

JavaApplication17.java | JavaApplication18.java | JavaApplication19.java

```java
public static void main(String[] args) {
    int x[][]={
        {1,2,3},
        {5,6,7},
        {9,8,4},
    };
    int max=x[0][0];
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++){
        if(max<x[i][j])
            max=x[i][j];
    System.out.print(x[i][j]+" ");
    }
    System.out.println();
    }
    System.out.println("max:"+max);
```

Output - JavaApplication19 (run)

```
run:
1 2 3
5 6 7
9 8 4
max:9
BUILD SUCCESSFUL (total time: 0 seconds)
```

طباعة عناصر مصفوفة ثنائية البعد مع ايجاد الرقم الأكبر

JavaApplication19 - NetBeans IDE 8.2

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help

JavaApplication17.java | JavaApplication18.java | JavaApplication19.java

```java
public static void main(String[] args) {
    int x[][]={
        {1,2,3},
        {5,6,7},
        {9,8,4},
    };
    int min=x[0][0];
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++){
        if(min>x[i][j])
            min=x[i][j];
    System.out.print(x[i][j]+" ");
    }
    System.out.println();
    }
    System.out.println("min:"+min);
```

Output - JavaApplication19 (run)

```
run:
1 2 3
5 6 7
9 8 4
min:1
BUILD SUCCESSFUL (total time: 0 seconds)
```

طباعة عناصر مصفوفة ثنائية البعد مع ايجاد الرقم الأصغر

JavaApplication19 - NetBeans IDE 8.2

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help

```java
package javaapplication19;
public class JavaApplication19 {
public static void main(String[] args) {
    int x[][]={
        {1,2,3},
        {5,6,7},
        {9,8,4},
    };
    int min=x[0][0],max=x[0][0];
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++){
        if(min>x[i][j])
            min=x[i][j];
        if(max<x[i][j])
            max=x[i][j];
    System.out.print(x[i][j]+" ");
    }
    System.out.println();
    }
    System.out.println("max:"+max);
    System.out.println("min:"+min);
```

Output - JavaApplication19 (run)

```
run:
1 2 3
5 6 7
9 8 4
max:9
min:1
BUILD SUCCESSFUL (total time: 0 seconds)
```

طباعة عناصر مصفوفة ثنائية البعد مع ايجاد الرقم الأكبر والرقم الأصغر

JavaApplication19 - NetBeans IDE 8.2

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help

JavaApplication17.java | JavaApplication18.java | JavaApplication19.java

```java
package javaapplication19;
public class JavaApplication19 {
public static void main(String[] args) {
    int x[][]={
        {1,2,3},
        {5,6,7},
        {9,8,4},
    };
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++)
    System.out.print(x[i][j]+" ");
     System.out.println();
    }
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++)
    if(i==j)
    System.out.println(x[i][j]+"");
     else
     System.out.print(" ");
     System.out.println("");
    }}}
```

Output - JavaApplication19 (run)

```
run:
1 2 3
5 6 7
9 8 4
1

 6

  4

BUILD SUCCESSFUL (total time: 0 seconds)
```

ولا ننسى الشرط للقطر الرئيسي هو i==j

طباعة عناصر مصفوفة ثنائية البعد مع طباعة القطر الرئيسي فقط

JavaApplication19 - NetBeans IDE 8.2

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help

JavaApplication17.java | JavaApplication18.java | JavaApplication19.java

```java
package javaapplication19;
public class JavaApplication19 {
public static void main(String[] args) {
    int x[][]={
        {1,2,3},
        {5,6,7},
        {9,8,4},
    };
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++)
    System.out.print(x[i][j]+" ");
     System.out.println();
    }
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++)
    if(i+j==2)
    System.out.println(x[i][j]+"");
     else
     System.out.print(" ");
     System.out.println("");
    }}}
```

Output - JavaApplication19 (run)

```
run:
1 2 3
5 6 7
9 8 4
  3

 6

9

BUILD SUCCESSFUL (total time: 0 seconds)
```

ولا ننسى الشرط للقطر الثانوي هو i+j==n-1

حيث ال n تمثل عدد الصفوف أو عدد الأعمدة

طباعة عناصر مصفوفة ثنائية البعد مع طباعة القطر الثانوي فقط

JavaApplication19 - NetBeans IDE 8.2

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help

JavaApplication17.java | JavaApplication18.java | JavaApplication19.java

```java
package javaapplication19;
public class JavaApplication19 {
public static void main(String[] args) {
    int x[][]={
        {1,2,3},
        {5,6,7},
        {9,8,4},
    };
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++)
    System.out.print(x[i][j]+" ");
     System.out.println();
    }
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++)
    if(x[i][j]%2==0)
    System.out.println("even: "+x[i][j]+" ");
    }}}
```

Output - JavaApplication19 (run)

```
run:
1 2 3
5 6 7
9 8 4
even: 2
even: 6
even: 8
even: 4
BUILD SUCCESSFUL (total time: 0 seconds)
```

طباعة عناصر مصفوفة ثنائية البعد مع طباعة العناصر الزوجية

JavaApplication19 - NetBeans IDE 8.2

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help

JavaApplication17.java | JavaApplication18.java | JavaApplication19.java

```java
package javaapplication19;
public class JavaApplication19 {
public static void main(String[] args) {
    int x[][]={
        {1,2,3},
        {5,6,7},
        {9,8,4},
    };
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++)
    System.out.print(x[i][j]+" ");
     System.out.println();
    }
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++)
    if(x[i][j]%2!=0)
    System.out.println("odd: "+x[i][j]+" ");
    }}}
```

Output - JavaApplication19 (run)

```
run:
1 2 3
5 6 7
9 8 4
odd: 1
odd: 3
odd: 5
odd: 7
odd: 9
BUILD SUCCESSFUL (total time: 0 seconds)
```

طباعة عناصر مصفوفة ثنائية البعد مع طباعة العناصر الفردية

JavaApplication19 - NetBeans IDE 8.2

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help

JavaApplication17.java | JavaApplication18.java | JavaApplication19.java

```java
package javaapplication19;
public class JavaApplication19 {
public static void main(String[] args) {
    int x[][]={
        {1,2,3},
        {5,6,7},
        {9,8,-4},
    };
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++)
    System.out.print(x[i][j]+" ");
     System.out.println();
    }
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++)
    if(x[i][j]>0)
    System.out.println("positive: "+x[i][j]+" ");
    }}}
```

Output - JavaApplication19 (run)

```
run:
1 2 3
5 6 7
9 8 -4
positive: 1
positive: 2
positive: 3
positive: 5
positive: 6
positive: 7
positive: 9
positive: 8
BUILD SUCCESSFUL (total time: 0 seconds)
```

طباعة عناصر مصفوفة ثنائية البعد مع طباعة العناصر الموجبة

حل واجبات الجزء الأول
المصفوفة ثنائية البعد
2D array

JavaApplication31 - NetBeans IDE 8.2

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help

Start Page | JavaApplication31.java

Source | History

```java
public static void main(String[] args) {
    {
        Scanner sc=new Scanner(System.in);
        int n,m;
        System.out.println("Enter the total number of elements ");
        System.out.println("Enter the row size");
        n=sc.nextInt();
        System.out.println("Enter the colum size");
        m=sc.nextInt();
        int arr[][]=new int[n][m];
        System.out.println("Enter the elements of the array ");
        for(int i=0; i<n ;i++)
        for(int j=0; j<m ;j++){

            arr[i][j]=sc.nextInt();
        }

        System.out.println("Enter the element whose frequency you want to know");
        int ele=sc.nextInt();
        int occ=0;
        for(int i=0;i<n;i++)
            for(int j=0;j<n;j++)
        {
            if(ele==arr[i][j])
            {
                occ++;
            }
        }
        System.out.println(ele+" occurred "+occ+" times ");}
}}
```

Output - JavaApplication31 (run)

```
run:
Enter the total number of elements
Enter the row size
4
Enter the colum size
4
Enter the elements of the array
1 2 3 4
2 3 4 5
1 2 3 4
1 2 3 5
Enter the element whose frequency you want to know
2
2 occurred 4 times
BUILD SUCCESSFUL (total time: 25 seconds)
```

طباعة عناصر مصفوفة ثنائية البعد مع طباعة العنصر المتكرر

JavaApplication19 - NetBeans IDE 8.2

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help

```java
package javaapplication19;
public class JavaApplication19 {
public static void main(String[] args) {
    int x[][]={
        {1,2,3},
        {5,6,7},
        {9,8,-4},
    };
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++)
    System.out.print(x[i][j]+" ");
     System.out.println();
    }
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++)
    if(x[i][j]<0)
    System.out.println("nagative: "+x[i][j]+" ");
    }}}
```

Output - JavaApplication19 (run)

```
run:
1 2 3
5 6 7
9 8 -4
nagative: -4
BUILD SUCCESSFUL (total time: 0 seconds)
```

طباعة عناصر مصفوفة ثنائية البعد مع طباعة العناصر السالبة

JavaApplication19 - NetBeans IDE 8.2

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help

JavaApplication17.java | JavaApplication18.java | JavaApplication19.java

```java
public class JavaApplication19 {
public static void main(String[] args) {
    int x[][]={
        {1,2,3},
        {5,6,7},
        {9,8,4},
    };
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++)
    System.out.print(x[i][j]+" ");
     System.out.println();
    }
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++)
    if(i==j)
    System.out.print(x[i][j]+" ");
    else if(i+j==2)
     System.out.print(x[i][j]);
     else
     System.out.print(" ");
     System.out.println(" ");
    }}}
```

Output - JavaApplication19 (run)

```
run:
1 2 3
5 6 7
9 8 4
1   3
 6
9 4
BUILD SUCCESSFUL (total time: 0 seconds)
```

طباعة عناصر مصفوفة ثنائية البعد مع طباعة القطر الرئيسي والقطر الثانوي

حل واجبات الجزء الثاني
المصفوفة ثنائية البعد
2D array

JavaApplication11 - NetBeans IDE 8.2

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help

Start Page | JavaApplication11.java

Source | History

```java
package javaapplication11;
public class JavaApplication11 {
    public static void main(String[] args) {
        int[][] a = { {1,2},{3,4} };
        int[][] b= { {2,0},{1,2}};
        int[][] y = new int[2][2];
        for(int i = 0; i < 2; i++) {
        for (int j = 0; j < 2; j++) {
        for (int k = 0; k <2; k++) {
        y[i][j] += a[i][k] * b[k][j];
        }}}
        System.out.println("Multiplication of two matrices is: ");
        for(int[] row : y) {
            for (int column : row) {
                System.out.print(column + "  ");
            }
                System.out.println();
            }}}
```

Output - JavaApplication11 (run)

```
run:
Multiplication of two matrices is:
4   4
10   8
BUILD SUCCESSFUL (total time: 0 seconds)
```

> Write down a Java program to find the multiplication of two 2D arrays?

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 1\times2+2\times1 & 1\times0+2\times2 \\ 3\times2+4\times1 & 3\times0+4\times2 \end{bmatrix} = \begin{bmatrix} 4 & 4 \\ 10 & 8 \end{bmatrix}$$

JavaApplication19 - NetBeans IDE 8.2

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help

Start Page | JavaApplication16.java | JavaApplication19.java

Source | History

```java
package javaapplication19;
public class JavaApplication19 {
public static void main(String[] args) {
    int x[]={2,3,4,5};
    for(int i=0;i<4;i++)
    System.out.println(x[i]+" ");
    for(int j=3;j>=0;j--)
    System.out.print(x[j]+" ");
    }}
```

Output - JavaApplication19 (run)

```
run:
2
3
4
5
5 4 3 2 BUILD SUCCESSFUL (total time: 0 seconds)
```

طباعة عناصر مصفوفة أحادية بالعكس

Write down a Java program to reverse the order of elements in an array?
for example: arr1 = {12, 30, 15, -4, 7} becomes arr1= {7, -4, 15, 30, 12}

حل أسئلة سنوات سابقة
على المصفوفات

JavaApplication19 - NetBeans IDE 8.2

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help

JavaApplication17.java | JavaApplication18.java | JavaApplication19.java

```java
package javaapplication19;
public class JavaApplication19 {
public static void main(String[] args) {
    int x[][]={
        {1,2,3},
        {5,6,7},
        {9,8,4},
    };
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++)
    System.out.print(x[i][j]+" ");
    System.out.println();
    }
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++)
    if(x[i][j]>0 &&  i==j )
    System.out.println("positive: "+x[i][j]+" ");
    else if(x[i][j]>0 && i+j==2)
    System.out.println("positive: "+x[i][j]+" ");
    }}}
```

Output - JavaApplication19 (run)

```
run:
1 2 3
5 6 7
9 8 4
positive: 1
positive: 3
positive: 6
positive: 9
positive: 4
BUILD SUCCESSFUL (total time: 0 seconds)
```

طباعة عناصر مصفوفة ثنائية البعد مع طباعة العناصر الموجبة فقط من القطر الرئيسي والقطر الثانوي

JavaApplication19 - NetBeans IDE 8.2

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help

Start Page | JavaApplication16.java | JavaApplication19.java

Source | History

```java
package javaapplication19;
public class JavaApplication19 {
public static void main(String[] args) {
    int x[]={2,3,4,5};
    for(int i=0;i<4;i++)
    System.out.println(x[i]+" ");
    for(int j=3;j>=0;j--)
    System.out.print(x[j]+" ");
    }}
```

Output - JavaApplication19 (run)

```
run:
2
3
4
5
5 4 3 2 BUILD SUCCESSFUL (total time: 0 seconds)
```

طباعة عناصر مصفوفة أحادية بالعكس

Write down a Java program to reverse the order of elements in an array?
for example: arr1 = {12, 30, 15, -4, 7} becomes arr1= {7, -4, 15, 30, 12}

أسئلة أكثر تطوير

JavaApplication19 - NetBeans IDE 8.2

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help

Start Page | JavaApplication16.java | JavaApplication19.java

Source | History

```java
package javaapplication19;
public class JavaApplication19 {
public static void main(String[] args) {
    int x[][]={
        {-1,2,3},
        {5,-6,7},
        {-9,8,-4},
    };
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++)
    System.out.print(x[i][j]+" ");
    System.out.println();
    }
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++)
    if(x[i][j]<0 &&  i==j )
    System.out.println("negateve: "+x[i][j]+" ");
    else if(x[i][j]<0 && i+j==2)
    System.out.println("negateve: "+x[i][j]+" ");
    }}}
```

Output - JavaApplication19 (run)

```
run:
-1 2 3
5 -6 7
-9 8 -4
negateve: -1
negateve: -6
negateve: -9
negateve: -4
BUILD SUCCESSFUL (total time: 0 seconds)
```

طباعة عناصر مصفوفة ثنائية البعد مع طباعة العناصر السالبة فقط من القطر الرئيسي والقطر الثانوي

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JavaApplication19 - NetBeans IDE 8.2

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help

Start Page | JavaApplication16.java | JavaApplication19.java

Source | History

```java
package javaapplication19;
public class JavaApplication19 {
public static void main(String[] args) {
    int x[][]={
        {1,2,3},
        {5,6,7},
        {9,8,4},
    };
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++)
    System.out.print(x[i][j]+" ");
    System.out.println();
    }
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++)
    if(x[i][j]%2==0 &&  i==j )
    System.out.println("even: "+x[i][j]+" ");
    else if(x[i][j]%2==0 && i+j==2)
    System.out.println("even: "+x[i][j]+" ");
    }}}
```

Output - JavaApplication19 (run)

```
run:
1 2 3
5 6 7
9 8 4
even: 6
even: 4
BUILD SUCCESSFUL (total time: 0 seconds)
```

طباعة عناصر مصفوفة ثنائية البعد مع طباعة العناصر الزوجية فقط من القطر الرئيسي والقطر الثانوي

JavaApplication19 - NetBeans IDE 8.2

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help

Start Page | JavaApplication16.java | JavaApplication19.java

Source History

```java
package javaapplication19;
public class JavaApplication19 {
public static void main(String[] args) {
    int x[][]={
        {1,2,3},
        {5,6,7},
        {9,8,4},
    };
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++)
    System.out.print(x[i][j]+" ");
    System.out.println();
    }
    for(int i=0;i<3;i++){
    for(int j=0;j<3;j++)
    if(x[i][j]%2!=0 &&  i==j )
    System.out.println("odd: "+x[i][j]+" ");
    else if(x[i][j]%2!=0 && i+j==2)
    System.out.println("odd: "+x[i][j]+" ");
    }}}
```

Output - JavaApplication19 (run)

```
run:
1 2 3
5 6 7
9 8 4
odd: 1
odd: 3
odd: 9
BUILD SUCCESSFUL (total time: 0 seconds)
```

طباعة عناصر مصفوفة ثنائية البعد مع طباعة العناصر الفردية فقط من القطر الرئيسي والقطر الثانوي